

- 2 The interpupillary distance (IPD) is the distance in millimetres between the centres of the pupils of the eyes. Fig. 2.1 shows how IPD is measured.

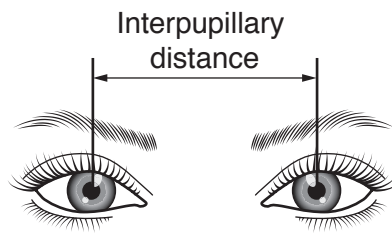


Fig. 2.1

IPD is one example of a characteristic of human facial structure that shows variation.

Fig. 2.2 shows the pattern of variation in IPD in a large sample of adults.

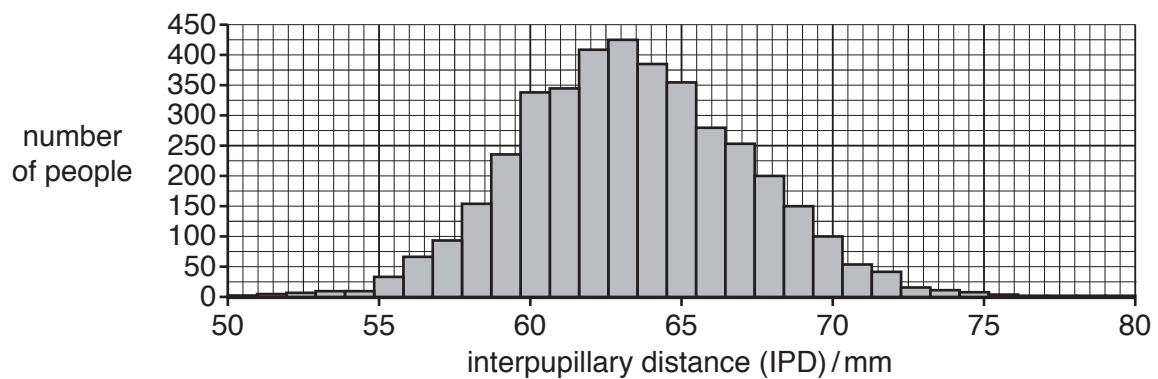


Fig. 2.2

- (a) (i) Name the type of variation shown in Fig. 2.2.

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- (ii) Suggest **and** explain how genes and the environment contribute to variation in IPD in humans.

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- (b)** Individuals with an IPD of 70 mm or more have a mutation in the *PAX3* gene that results in less PAX3 protein being made.

The normal role of the PAX3 protein is to increase the expression of many other genes involved in embryonic development. These genes affect a range of phenotypic features such as facial structure, hearing and eye colour.

- (i) State the term that is used to describe a gene, such as *PAX3*, that controls the expression of other genes **and** suggest how the *PAX3* protein controls the expression of other genes.

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- (ii) Describe how microarray analysis could be used to identify the genes switched on by PAX3 in embryonic cells.

[5]

- (iii) The chimpanzee, *Pan troglodytes*, has DNA that is 98.5% similar to humans, including possession of the *PAX3* gene. Investigations show that chimpanzees express higher levels of the PAX3 protein during embryonic development than humans.

Fig. 2.3 shows a chimpanzee, *Pan troglodytes*.



Fig. 2.3

Suggest how knowledge of the *PAX3* gene helps scientists explain how humans and chimpanzees are very different in facial structure, even though they have very similar DNA.

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[Total: 15]